

Velī — Frontend Design System

Cloud Parallax (Pvt) Ltd · v0.1 · 2026-05-01

1 Purpose and audience

This document is the canonical reference for visual and interaction patterns used in the Velī smartphone surface (Tamil-first PWA, Android wrap via Capacitor). It is read by every contributor adding a page or component, and it is binding: pages must reach for the primitives defined here rather than inline raw Tailwind classes for things the primitives already cover.

Velī is operated by Cloud Parallax (Pvt) Ltd for users in Sri Lanka’s Northern Province. Three constraints shape every decision:

- **Tamil first.** The display register is colloquial Jaffna Tamil, not literary Tamil and not Indian Tamil. Tamil messages are authored first; English and Sinhala follow as translations.
- **Mobile first, low-end Android.** The PWA targets intermittent 3G, modest CPUs, and small screens. Touch targets must be reachable one-handed; transitions must not be the only signal.
- **Honest tier labelling.** Every document or claim shows its verification tier explicitly in both English and Tamil. The platform never inflates a self-asserted artifact into something it is not. The tier values `self_asserted`, `community_corroborated`, `authority_attested` are normative and never paraphrased.

2 Tokens

2.1 Colour

The base palette is the shadcn-svelte zinc scale in OKLCH, declared in `frontend/src/app.css`. Tailwind utility classes resolve to these CSS custom properties; **never** hex-code colours in templates.

Token	Tailwind class	Use
background	<code>bg-background</code>	Page canvas behind everything.
foreground	<code>text-foreground</code>	Default body text.
primary	<code>bg-primary</code> / <code>text-primary</code>	Headlines, key actions, the Tamil hero word.
secondary	<code>bg-secondary</code> / <code>text-secondary-foreground</code>	Inactive toggles, neutral pills.
muted	<code>bg-muted</code> / <code>text-muted-foreground</code>	De-emphasised supporting text and surfaces.
accent	<code>bg-accent</code>	Hover and focus surfaces.
destructive	<code>bg-destructive</code>	Error states, “backend unreachable”, invalid input.
card	<code>bg-card</code>	Raised surfaces (cards, dialogs, status strips).
border	<code>border</code> / <code>border-border</code>	Default 1 px hairline.
ring	<code>ring-ring</code>	Focus halos at 2 px width with 2 px offset.

Dark mode swaps the same tokens; never hard-code light-mode values.

2.2 Typography

Display register is Tamil. The system font stack must include a Northern Province–suitable Tamil font (Inter, Noto Sans Tamil, or the device default). The Tailwind type scale collapses to four levels:

Role	Class	Notes
Hero	text-7xl sm:text-8xl font-bold	Reserved for the Tamil wordmark “வெளி” on the landing surface only.
Page title	text-xl font-semibold	<h1> on every page below the layout.
Section	text-lg font-semibold	<h2> inside cards.
Body	text-sm or text-base	Default copy. Use text-base sm:text-lg for long-form prose.
Caption	text-xs text-muted-foreground	Tier labels, badges, footnotes.

Always pair Tamil and English: never replace a Tamil label with an English-only one in user-facing surfaces. The IVR/USSD/SMS surfaces follow their own register rules and are out of scope here.

2.3 Spacing and radius

Tailwind’s default scale; preferred increments are 2 / 3 / 4 / 6 / 8 / 12 / 16 (0.5 rem multiples). Never invent custom pixel values when a token will do.

Radius tokens come from --radius: 0.625rem:

- rounded-sm — pills, badges
- rounded-md — buttons, inputs, status strips
- rounded-lg — cards, dialog surfaces
- rounded-xl — modal overlays, hero containers

Borders are 1 px (border) on border-border. Avoid heavier weights unless you are crossing a destructive or focus boundary.

3 Components

Every primitive lives at frontend/src/lib/components/ui/<name>.svelte and is re-exported from frontend/src/lib/components/ui/index.ts. Pages import the primitive, never the underlying Tailwind elements when a primitive applies.

Primitives use Svelte 5 runes (\$props, \$state, \$derived, \$effect) and the cn(...) helper from \$lib/utils for class composition with tailwind-merge.

3.1 Button

```
import Button from '$lib/components/ui/Button.svelte'
```

Variants: primary (default), secondary, ghost, destructive. Sizes: sm, md (default), lg. Touch target is 44 px minimum at md to satisfy WCAG 2.5.5 on phones.

```
<Button variant="primary" onclick={handleSubmit}>{m.submit()}</Button>
```

3.2 Card

Surface for grouped content: a header (title + optional badge), an optional content body, and an optional footer. The current landing page’s three-phase strip is the canonical example.

```
<Card>
  <CardHeader title={m.phase1_title()} />
  <CardContent>{m.phase1_blurb()}</CardContent>
</Card>
```

3.3 Badge

Inline pill for short states (locale, role, count). Variants: `default`, `secondary`, `destructive`. Use `Badge` for generic state; use `TierBadge` (below) for verification tiers — never mix them.

3.4 TierBadge

A specialised `Badge` for the three-tier verification model. It **always** renders both English and Tamil labels per the architectural contract in the strategic plan §Phase 0. Visual treatment makes the hierarchy of legal effect obvious without saying so:

Tier value	EN label	TA label	Visual
<code>self_asserted</code>	Self-asserted	சுய அறிவிப்பு	<code>bg-muted text-muted-foreground</code> — visibly the lightest.
<code>community_corroborated</code>	Community-corroborated	சமூக சாட்சியம்	<code>bg-secondary text-secondary-foreground</code> — neutral, no claim of legal effect.
<code>authority_attested</code>	Authority-attested	அதிகாரப்பூர்வ உறுதி	<code>bg-primary text-primary-foreground</code> — the most confident treatment, used only when the tier carries legal effect.

Never substitute tier names. Never add a fourth tier, even visually. If a UI surface needs to express a derived state (e.g. “expired attestation”), do it with a separate adjacent badge, not by changing the tier label.

3.5 StatusStrip

Thin bar at the bottom of a page (or pinned to the layout) reporting system reachability. Two states:

- `reachable` — `bg-card text-muted-foreground`, with an optional phase-number `Badge` from the backend hello payload.
- `unreachable` — `bg-destructive text-destructive-foreground`, plus the parsed error string at reduced opacity for dev visibility.

The current landing page’s status strip is the reference; primitives should preserve its exact visual.

3.6 LocaleSwitcher

Three-button group (தமிழ் / English / 🇩🇪🇩🇪🇩🇪) calling `Paraglide’s setLocale`. Active locale uses `bg-primary text-primary-foreground`; inactive uses `bg-secondary text-secondary-foreground`. Mounted at fixed `top-4 right-4 z-50` in the layout. Buttons are 44 px tall.

4 Patterns

4.1 Page shell

Every page wraps content in:

```
<div class="flex min-h-screen flex-col bg-background text-foreground">
  <main class="mx-auto w-full max-w-2xl px-6 py-12">
    <!-- page body -->
  </main>
</div>
```

The landing page widens to `max-w-5xl`. The user-detail page uses the default `max-w-2xl`. Avoid full-bleed layouts on phones.

4.2 List of properties

Definition list (`<dl>` / `<dt>` / `<dd>`) divided by hairlines is the canonical pattern for displaying record fields (e.g. user details). Each row is `flex items-baseline gap-4 px-6 py-3`; the `<dt>` is `w-36 shrink-0 text-sm font-medium text-muted-foreground`; the `<dd>` is `text-sm`. Monospace (`font-mono`) for IDs and NIC numbers.

4.3 Error and empty states

Localised, never English-only. Use a Card with centred copy:

```
<Card class="text-center">
  <CardContent>{m.user_not_found()}</CardContent>
</Card>
```

Server errors get `border-destructive/30 bg-destructive/10` plus a single line of guidance (“please try again later” — never a stack trace). Never expose raw backend error envelopes to the user; log them.

4.4 Form inputs

All inputs are full-width on mobile, 44 px tall, with a label above (never placeholder-only — accessibility regression on iOS Safari and some Tamil IMEs). Validation errors surface below the input in `text-sm text-destructive`.

5 Accessibility

- **Touch targets:** 44 px × 44 px minimum. Buttons at `size="md"` satisfy this; never go below `size="sm"` for primary actions.
- **Focus rings:** never disable. The `ring-ring` token gives a 2 px halo at 2 px offset; tab order must be logical.
- **Contrast:** zinc OKLCH already meets AA at body sizes; verify manually for tier-coloured backgrounds.
- **aria-label:** every interactive control needs one, **and it must be localised through Paraglide**. Never hard-code English `aria-label` strings.
- **Language tag:** the `<html lang>` is set by Paraglide via the `%paraglide.lang%` placeholder; do not override it manually.
- **Right-to-left:** not currently in scope. Tamil and Sinhala are LTR.

6 Internationalisation rules

- **Tamil first.** Author the Tamil string before any other language. Never translate Tamil from English in user-facing copy.
- **Register.** Northern Province colloquial (Jaffna). Avoid literary Tamil. Have a Northern Province native speaker review new strings before merge.
- **Catalogue layout.** All keys in `frontend/messages/{en,ta,si}.json`, one key per UI string. No string concatenation across keys — pluralisation and parameter substitution use Paraglide’s parameter syntax.
- **Numerals.** Use European numerals (0–9) by default; the platform shows Tamil numerals only when explicitly requested by content.
- **Quoting.** Tamil uses straight quotes. Sinhala uses standard punctuation. English may use curly quotes in long-form copy but not in UI labels.

7 Mobile-first defaults

- Default breakpoints follow Tailwind: `sm` 640 px, `md` 768 px, `lg` 1024 px. The smartphone surface targets `< sm`; widen up from there.
- Never assume hover. Every interactive state must work on touch.
- Service worker caches the active session; pages should render meaningful content with cached data plus a small banner when offline. Avoid spinner-only offline states.
- Capacitor Android wrap: status-bar colour is `bg-background`; the splash screen is hidden on layout mount.

8 Code conventions

- **Svelte 5 runes only.** `$state`, `$derived`, `$effect`, `$props`. No legacy writable or readable stores.
- **Server-only code** lives in `src/lib/server/`. Never import from `$lib/server/` inside a `.svelte` component.
- **API calls** go through `+page.server.ts load()` using the typed client at `$lib/api`. Never `fetch()` the backend directly from a component.
- **Primitive imports.** Reach for `$lib/components/ui/<name>.svelte` before writing new Tailwind classes for buttons, cards, badges, or status strips.
- **`cn()` helper.** Compose classes with `cn(...)` from `$lib/utils`. Never string-concatenate Tailwind classes.
- **Props.** Type props with TypeScript interfaces; default values set via destructuring at the `$props()` site. Forward `class` and `ref` through `WithoutChildrenOrChild<T>` from `$lib/utils` when wrapping native elements.
- **No inline `<style>`** unless scoped and unavoidable. Tailwind utilities first; CSS custom properties when a token is needed.

9 Boundaries

This design system is for the smartphone surface (PWA + Capacitor Android wrap). It does **not** cover:

- The IVR / USSD / SMS surfaces (governed by Tamil voice prompt authoring rules, see strategic plan §Phase 0 architecture).
- Internal admin or back-office tooling (out of scope for v1.x).
- Print or PDF artefacts (deed-ready bundles in Phase 2 use a separate Typst template).

When in doubt, defer to `CLAUDE.md` for project-wide rules and the strategic plan in `docs/plan/` for product policy.